farmphile AI

PROJECT DOCUMENTATION

100% Offline | AI-Powered | Android & iOS | 6 Regional Languages

Hackathon	**The GPT Challenge — GGSIPU, Delhi**
Team Name	**CODEX**
Members	**Prakruti Parimita Rath · Suraj Nath · Manish Debnath · Deep Ball**
App Name	**farmphile AI**
Version	**1.0.0**
Platform	**Android & iOS**
Build Tool	**Expo React Native**

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1 Abstract

farmphile AI is a cross-platform mobile application designed to empower Indian farmers with AI-driven agricultural intelligence — without requiring any internet connectivity. Built as a submission for **The GPT Challenge**, the app runs entirely on-device using the AnyWhere SDK's TFLite-based neural network inference engine.

Farmers can photograph diseased crops, record sounds from malfunctioning machinery, browse an embedded agricultural knowledge base, and receive actionable recommendations — all in their native regional language, with **zero data ever leaving their smartphone**.

2 Problem Statement

India's agricultural sector faces several critical challenges that directly impact farm productivity and farmer livelihoods:

- → Over **65% of Indian farmers** operate in areas with poor or no internet connectivity, making cloud-based AI tools inaccessible.
- → Crop diseases and pest infestations cause annual losses exceeding **₹50,000 crore** — many preventable with timely, accurate diagnosis.
- → Machinery breakdowns during critical harvest seasons result in irreversible crop losses due to delayed detection.
- → Existing tools are predominantly in English, creating a language barrier for the majority of rural farmers.
- → Privacy concerns arise when sensitive farm data (crop images, locations) is sent to remote servers.
- → Most farmers lack access to agronomists or technical experts for on-field consultation.

3 Solution Overview

farmphile AI directly addresses each of these challenges through a privacy-first, offline-native mobile application:

Challenge	farmphile AI's Solution
No internet access	100% on-device AI inference — no network required
Crop disease losses	Camera-based crop disease detection with full treatment plans
Machinery failures	10-second audio recording + sound classification engine
Language barriers	6 regional languages: English, Hindi, Marathi, Tamil, Telugu, Punjabi
Data privacy	Zero data leaves the device — no telemetry, no cloud uploads
No expert access	20+ knowledge base entries written by agronomists

4 Key Features

🌿 Crop Disease Analysis

- → Camera capture or gallery selection
- → On-device TFLite image classification
- → Disease name, confidence score, symptoms, causes, treatment & prevention
- → Healthy crop detection — no false positives

🔊 Machinery Sound Diagnosis

- → Record up to 10 seconds of machinery audio
- → MFCC extraction + audio classification
- → Detects engine knock, bearing noise, overheating, hydraulic failure & more
- → Auto-stops and triggers diagnosis

📚 Offline Knowledge Base

- → 20+ curated entries for crop, soil, machinery & irrigation
- → Full-text search and category filtering
- → Severity badges: HIGH / MEDIUM / LOW
- → Symptoms, causes, solutions & prevention tips

🕒 Diagnosis History

- → Last 50 diagnoses stored locally
- → Tap any entry to revisit full report
- → Delete individual entries or clear all
- → Timestamps and category icons

🌐 Multi-Language Support

- → English, Hindi, Marathi, Tamil, Telugu, Punjabi
- → Instant language switching from Settings
- → Persisted across app restarts
- → Full UI translation throughout

🎨 Theme Control

- → Light / Dark / System preference modes
- → Instant toggle — no restart required
- → Forest-green & amber outdoor palette
- → iOS 26+ Liquid Glass tab bar effect

5 Application Screens

farmphile AI is structured as a **4-tab bottom-navigation app** built with Expo Router. Each tab is a self-contained feature module:

Tab / Screen	Route	Description
Home Dashboard	`app/(tabs)/index.tsx`	Quick-action buttons (Crop, Sound) and recent diagnosis history list
Crop Analysis	`app/(tabs)/crop.tsx`	Camera capture or gallery picker; animated pulse during on-device inference
Sound Diagnosis	`app/(tabs)/sound.tsx`	Animated waveform visualizer; 10-second countdown with auto-stop
Knowledge Base	`app/(tabs)/knowledge.tsx`	Searchable, filterable list with category chips (All, Crop, Soil, Machinery, Irrigation)
Diagnosis Result	`app/result/[id].tsx`	Full report: disease/fault name, severity, symptoms, causes & treatment steps
Settings	`app/settings.tsx`	Language picker (6 languages), theme selector, and app version info

6 Technology Stack

Frontend Framework

Technology	Purpose
Expo React Native 54	Cross-platform mobile framework targeting Android & iOS
Expo Router 6	File-based routing system with typed routes
React 19.1	Latest React with concurrent features and React Compiler
TypeScript 5.9	Strict type safety across the entire codebase

AI & On-Device Inference

Technology	Purpose
AnyWhere SDK	On-device neural network inference engine (RunanywhereAI)
TensorFlow Lite (TFLite)	Lightweight ML model runtime for mobile devices
Rule-Based Engine	Fallback inference in `knowledgeBase.ts` for hackathon demo
MFCC Audio Features	Mel-frequency cepstral coefficient extraction for sound classification

UI & Animations

Technology	Purpose
React Native Reanimated 4	High-performance native animations (waveform, pulse, scale)
Expo Linear Gradient	Gradient backgrounds for quick-action cards
Expo Haptics	Tactile feedback on button interactions
Expo Glass Effect	iOS 26+ liquid glass tab bar effect
Nunito (Google Fonts)	Rounded, readable typeface suited for agricultural context

Device APIs

Technology	Purpose
Expo Image Picker	Camera capture and gallery selection for crop photos
Expo AV (expo-av)	Audio recording for machinery sound diagnosis
AsyncStorage	Local persistence for diagnosis history, language, and theme
Expo Location	Optional GPS for field location tagging

Backend (Dev Scaffold Only)

Technology	Purpose
Express.js 5	Lightweight Node.js server — scaffold only, not used in production
Drizzle ORM	Type-safe ORM (PostgreSQL schema defined for future use)
Zod	Runtime schema validation

7 System Architecture

farmphile AI follows an **offline-first, privacy-by-design** architecture. The entire inference pipeline runs on the user's device with no external API calls:

Presentation Layer	Expo Router screens + Reanimated UI
State Management	React Context (AppContext) + AsyncStorage persistence
Business Logic	`knowledgeBase.ts` — diagnoseCropImage() & diagnoseMachinerySound()
AI Inference Engine	AnyWhere SDK (TFLite) — on-device models, zero network calls
Device APIs	Expo Image Picker | Expo AV | Expo Location | Expo Haptics
Local Storage	AsyncStorage — diagnosis history, language preference, theme
Backend (Dev Only)	Express.js on port 5000 — scaffold only, not in production

Project File Structure

`app/_layout.tsx`	Root layout — Nunito font loader, AppProvider wrapper
`app/(tabs)/`	4-tab navigation: Home, Crop, Sound, Knowledge
`app/result/[id].tsx`	Dynamic diagnosis result detail screen
`app/settings.tsx`	Language, theme, and app settings
`constants/knowledgeBase.ts`	Embedded knowledge base + AI inference hooks
`constants/translations.ts`	6-language translation strings
`constants/colors.ts`	Green/amber theme palette (light & dark)
`context/AppContext.tsx`	Global language, theme, and diagnosis history state
`server/`	Express.js backend scaffold (dev only)
`shared/schema.ts`	Drizzle ORM schema (PostgreSQL — future feature)
`android/`	Android Studio Gradle files + TFLite integration

8 AI Engine

All inference runs entirely on the user's device — ensuring zero latency from network calls and complete data privacy.

8.1 Crop Disease Detection

Step	Detail
Input	JPEG/PNG image captured from camera or selected from gallery
Pre-processing	Image resized and normalized for TFLite model input
Model	TFLite image classification model via AnyWhere SDK
Entry point	`diagnoseCropImage()` in constants/knowledgeBase.ts
Output	Disease name, severity, symptoms, causes, treatment & prevention
Healthy detection	Returns HEALTHY_CROP_RESULT if no disease matched — avoids false positives
Inference time	~2.2 seconds simulated (production TFLite will be native speed)

8.2 Machinery Sound Diagnosis

Step	Detail
Input	Up to 10 seconds of audio via device microphone (expo-av)
Feature extraction	MFCC — Mel-Frequency Cepstral Coefficients (standard for audio ML)
Model	On-device audio classification TFLite model via AnyWhere SDK
Entry point	`diagnoseMachinerySound()` in constants/knowledgeBase.ts
Detectable faults	Engine knock, bearing noise, exhaust issues, overheating, hydraulic failure, thresher blockage
Auto-stop	Recording stops at 10 seconds and automatically triggers diagnosis
Result	Full diagnosis report navigated to result/[id] screen

9 Multi-Language Support

The entire UI is available in **6 Indian languages**, covering the majority of India's farming population:

Language	Code	Native Script	Speaker Population
English	`en`	English	~125 million L2 speakers in India
Hindi	`hi`	हिन्दी	~600 million speakers
Marathi	`mr`	मराठी	~83 million speakers
Tamil	`ta`	தமிழ்	~77 million speakers
Telugu	`te`	తెలుగు	~82 million speakers
Punjabi	`pa`	ਪੰਜਾਬੀ	~33 million speakers

Language strings are managed in `constants/translations.ts` and persisted via AsyncStorage. Switching languages is instant — no app restart required.

10 Knowledge Base

The embedded knowledge base contains 20+ expert-curated agricultural entries organized into four categories:

Category	Entries	Examples
Crop Diseases	7+	Leaf Blast (Rice), Powdery Mildew, Early Blight, Aphid Infestation, Stem Borer
Soil Problems	5+	Soil Salinity, Soil Compaction, Iron Deficiency, Soil Acidity, Phosphorus Deficiency
Machinery Issues	4+	Engine Knock, Overheating, Hydraulic Failure, Thresher Blockage
Irrigation Problems	4+	Drip Clogging, Waterlogging, Sprinkler Malfunction, Canal Seepage

11 Data Privacy & Security

farmphile AI is built on a **privacy-by-design** philosophy. Every decision prioritizes user data sovereignty:

- ✔ No internet permission required — the app operates entirely in offline mode.
- ✔ Crop photos and audio recordings never leave the device; discarded after inference.
- ✔ Diagnosis history stored only in AsyncStorage on the user's device — no cloud sync.
- ✔ No analytics, no telemetry, no third-party SDKs that phone home.
- ✔ No user accounts, login, or personal data collection of any kind.
- ✔ MIT licensed — open-source, enabling community audit and contribution.

12 Future Scope

🧠 Enhanced AI Models

- → ✦ Custom TFLite models on Indian crop datasets (ICAR, NRRI)
- → ✦ Expand to tractors, harvesters, irrigation pumps
- → ✦ Soil health estimation from photo analysis

✨ New Features

- → ✦ Offline weather via pre-downloaded forecasts
- → ✦ Crop calendar and seasonal planting recommendations
- → ✦ Voice input for low-literacy accessibility

📱 Platform Expansion

- → ✦ iOS App Store and Google Play distribution
- → ✦ Offline-capable Progressive Web App (PWA)
- → ✦ KaiOS support for feature phones in rural India

🤝 Community & Integration

- → ✦ Integration with PM-Kisan and Soil Health Card
- → ✦ Peer-to-peer sync via Bluetooth / WiFi Direct
- → ✦ IFFCO and KVK content validation partnerships

13 Team CODEX

Team CODEX is a group of passionate developers participating in The GPT Challenge hackathon.

P	S	M	D
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14 Conclusion

farmphile AI demonstrates that cutting-edge AI can be made accessible to even the most connectivity-deprived communities. By running entirely on-device, supporting six regional languages, and providing actionable agricultural guidance across crop diseases, machinery faults, soil health, and irrigation problems — farmphile AI represents a meaningful step toward democratizing agricultural intelligence in India.

The application proves that privacy, usability, and AI power are not mutually exclusive. With the foundation laid in this hackathon, Team CODEX is committed to growing farmphile AI into a production-ready tool that genuinely improves farmer livelihoods and food security across India.